a spur gear. In this gearing the worm will always be the driver. This gearing is used where a large speed reduction is desired. It is useful in indexing head, rotary table and in the feed rod of lathe.

9.6 GEAR TRAIN

Gear drive is used where moderate to large amount of power is to be transmitted at constant velocity ratio. If the driving gear rotates in the clockwise direction, the follower will rotate in the anti- clockwise direction. The velocity ratio of a gear drive depends on the number of teeth present on the driving gear and the driven gear.

VELOCITY RATIO OF GEAR DRIVE

Velocity ratio of a gear drive is the ratio of number of revolutions of driven shaft or driven gear to the number of revolutions of driving shaft or driving gear in a particular time.

If N_1 and N_2 are the number of revolution of driver and follower and T_1 and T_2 are the number of teeth on the driving gear and the driven gear.

$$\text{Velocity ratio} = \frac{N_2}{N_1} = \frac{T_1}{T_2}$$

EXAMPLE 5

If a gear having 48 teeth rotates at a speed of 600rpm. In clock-wise direction, what will be speed and direction of rotation of a gear having 72 teeth which is in mesh with the first one ?

T_1 = 48 teeth T_2 = 72 teeth

N_1 = 600 rpm N_2 = ?

Number of teeth on the driving gear (T_1) = 48 teeth

Number of teeth on the driven gear (T_2) = 72 teeth

Number of rotational of driving gear (N_1) = 600 rpm.

Number of rotational of driven gear (N_2) = ?

$$\frac{T_1}{T_2} = \frac{N_2}{N_1}$$

$$N_2 = \frac{T_1 \times N_1}{T_2}$$

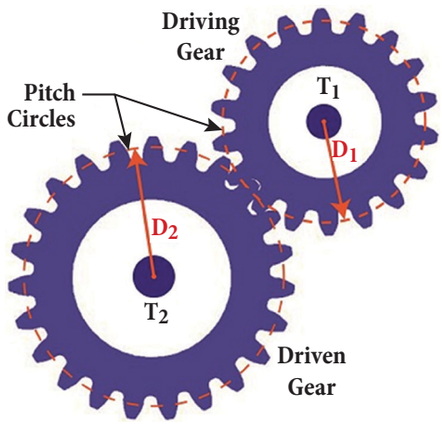
$$= \frac{48 \times 600}{72} = 400 \text{ rpm}$$

N_2 = 400 rpm Anti clock wise direction

SIMPLE GEAR TRAIN



If a gear train is arranged by keeping only one gear on a shaft, it is called simple gear train. The net velocity ratio of the gear drive is determined by the number of teeth present on the first and the last gears of the drive. The intermediate gears of the drive using only to fill the gap between the driving shaft and the driven shaft, is called



Simple Gear Train

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